

Dinesh Yadav

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PROFESSIONAL SUMMARY

AI Software Engineer with expertise building LLM-powered applications, scalable AI systems, RAG pipelines, agentic workflows, and production-ready backend architectures. Skilled in Generative AI, system design, and efficient algorithms, delivering reliable software at scale using Python FastAPI Docker Kubernetes AWS cloud services and observability.

EDUCATION

Indian Institute of Information Technology Vadodara

Nov 2022 - May 2026

B.Tech in Computer Science and Engineering

Relevant Coursework Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Machine Learning

WORK EXPERIENCE

BC2RI

May 2026 - Aug 2026

Software Developer

Remote

- Owned end-to-end frontend development of the company's MVP using React, TypeScript, Redux, and RTK Query, delivering a scalable healthcare platform supporting providers, patients, and caregivers.
- Designed role-based application architecture with protected routing, reusable UI components, and integration of 30+ REST APIs featuring centralized error handling, caching, and predictable state management.
- Implemented passwordless authentication using WebAuthn Passkeys and added core workflow features, including request approval/rejection systems and role-specific dashboards, enhancing security and user experience.

Tech Stack: React, TypeScript, RTK Query, REST APIs, WebAuthn

AeroYantra

Jun 2025 - May 2026

Software Engineer Intern

Noida

- Engineered 12+ interactive geospatial analysis tools using React and Cesium, enabling real-time distance, area, elevation, and spatial measurements that accelerated infrastructure planning and curbed dependence on manual site surveys.
- Created a multi-format geospatial visualization pipeline supporting PDF, DXF, KML/KMZ, and IFC overlays, and synchronized telemetry-driven drone simulations with video playback for accurate spatio-temporal infrastructure analysis.
- Refactored frontend map rendering by restructuring client-side workflows and caching base map resources, cutting API requests by 30% while adding secure authentication flows that lowered user support queries.

Tech Stack: React, Node.js, Express.js, MongoDB, Cesium.js

SapmenC

Feb 2025 - Jul 2025

Software Engineer Intern

Remote

- Built a collaborative project management platform featuring Kanban, List, Calendar, and WebSocket-based live messaging, streamlining task coordination and reducing manual team communication by 15+ hours per week.
- Optimized application performance using Next.js SSR, Zod schema validation, and Zustand state management, achieving 40% faster page loads and 25% higher query efficiency through seamless client-server data synchronization.

Tech Stack: Next.js, Zustand, Prisma, Node.js, MongoDB

PROJECTS

AgentLytics : Distributed Multi-Agent AI Platform for Intelligent Data Analysis

Live | GitHub

FastAPI, Redis, Docker, Python, ReactJS, Redux, SSE

- Architected a distributed AI data exploration platform using FastAPI, Redis, ARQ, Docker, and specialized agents, enabling conversational analysis of CSV, Excel, PDF, and TXT files with automated insights, visualizations, reports, and CSV exports in under 60 seconds.
- Slashed request routing latency from 3–4 seconds to under 200 ms with an ONNX-based intent classifier, dynamic agent routing, and a pre-warmed Docker sandbox pool that eliminated 4–5 second Python execution cold starts.
- Deployed isolated Document (RAG + ChromaDB) and Tabular (Docker sandbox) agents with dedicated memory, improving orchestration efficiency and response accuracy for complex analytical workflows.

Research Simplified : AI-Powered Research Paper Simplifier

Live | GitHub

LangChain, Redis Vector DB, Python, MongoDB, React

- Streamlined a low-latency research paper simplification pipeline through intelligent preprocessing, relevance-based chunk filtering, and parallel LLM inference, trimming LLM calls by 20% and delivering section-wise simplified explanations in 10–12 seconds.
- Delivered a production-grade RAG chatbot using Qdrant, cross-encoder re-ranking, and relevance classification, surfacing highly relevant context to produce grounded, context-aware responses in 2–3 seconds with improved retrieval accuracy.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++

AI Engineering: LangChain, LangGraph, Multi-Agent Systems, RAG, Vector Databases, LLMs, Prompt Engineering

Frontend Development: React, Next.js, Redux, RTK Query, Zustand, HTML5, CSS3

Backend Development: FastAPI, Node.js, Express.js, REST APIs, MongoDB, Redis, Prisma

Cloud, DevOps & Observability: Docker, Kubernetes, AWS, GitHub, GitLab, Prometheus, Grafana